

$$y' + 2\sin(2x) = 0$$

$$y' = -2\sin(2x)$$

$$y = \int -2\sin(2x) dx \rightarrow \int \sin(ax) dx = -\frac{1}{a} \cos(ax), a=2$$

$$y = -2(-\frac{1}{2} \cos(2x)) + C, y = \cos(2x) + C$$

$$y' = y, y = \int y dy = \frac{1}{2} y^2, y = Ce^x$$

$$y = ky \rightarrow y = Ce^{kx}$$

$$y' = -1.5y, k = -1.5 \rightarrow y = Ce^{-1.5x}$$

$$42 = 32 + 8 + 2 = 101010$$

0708

$$y' = \cos \pi x, \frac{dy}{dx} = \cos(\pi x) \quad \#9$$

$$y = \int \cos(\pi x) dx \rightarrow y = \frac{1}{\pi} \sin(\pi x) + C$$

#1

$$y' = 1 + y^2, (\frac{1}{4}\pi, 1)$$

$$y=0 \rightarrow y' = 1+0^2=1, y=1 \rightarrow y' = 1+1^2=2,$$

$$y=2 \rightarrow y' = 1+4=5$$

$$\frac{dy}{dx} = 1+y^2, \frac{dy}{1+y^2} = dx$$

$$\tan^{-1} y = x + C, y = \tan(x+C)$$

$$1 = \tan(\frac{\pi}{4} + C), \frac{\pi}{4} + C = \frac{\pi}{4}, C=0$$

$$\text{Hence } y = \tan x$$

#2

$$yy' + 4x = 0, (1,1), (0,2)$$

$$yy' = -4x, y \frac{dy}{dx} = -4x, y dy = -4x dx$$

$$\int y dy = \frac{1}{2} y^2, \int -4x dx = -2x^2$$

$$\frac{1}{2} y^2 = -2x^2 + C, y^2 = -4x^2 + C, y^2 + 4x^2 = C$$

$$i) (1,1)$$

$$1^2 + 4(1)^2 = C = 5$$

$$y^2 + 4x^2 = 5$$

$$y = \sqrt{5-4x^2}$$

$$ii) (0,2)$$

$$2^2 + 4(0)^2 = C = 4$$

$$y^2 + 4x^2 = 4$$

$$y = \sqrt{4-4x^2}$$

$\Rightarrow$  The two solutions are

#17

$$y' = y, y(0) = 1, h = 0.1$$

$$y_{n+1} = y_n + hf(x_n, y_n)$$

$$y_{n+1} = y_n + 0.1 y_n = 1.1 y_n$$

$$x_0 = 0, y_0 = 1$$

$$y_1 = 1.1 \times 1 = 1.1$$

$$y_2 = 1.1 \times 1.1 = 1.21$$

$$y_3 = 1.1 \times 1.21 = 1.331$$

$$y_4 = 1.1 \times 1.331 = 1.4641$$

$$y_5 = 1.1 \times 1.4641 = 1.61051$$

$$\Rightarrow y = e^x$$

#2

$$\text{Laplace transform of } y(t), \frac{dy}{dt} = y, y(0) = 2$$

$$\frac{dy}{dt} = y, y(0) = 2 \rightarrow \frac{dy}{y} = dt$$

$$\ln|y| = t + C, y = Ce^t$$

$$2 = Ce^0, C = 2 \Rightarrow y = 2e^t$$